Global Climatic Anomalies Associated with Extremes in the Southern Oscillation

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ABSTRACT

Composite temperature and precipitation anomalies during various stages of an event in the Southern Oscillation (SO) have been computed for several hundred stations across the globe. Large regions of coherent, significant signals are shown to exist for both extremes of the SO, with warm event signals generally opposite to those during cold events. In addition, during the year preceding the development of an event in the SO (year −1), climatic anomalies tend to be opposite to those during the following year (year 0). This confirms that the biennial tendency of the SO over the Pacific/Indian ocean sectors is also present in more remote regions with climatic signals related to the SO. Many of the signals are consistent enough from event to event to be useful for extended range forecasting purposes.

1. Introduction

The Southern Oscillation (SO) has been the subject of intensive investigation during the past 2 decades. This has been due in part to the increase in the quality and quantity of data over the tropical oceans, but also stems from the realization that the SO is intimately tied to planetary-scale climatic effects such as those observed in the intense 1982–83 event (Rasmusson and Wallace 1983; Philander 1983; Quiroz 1983; Kiladis and Diaz 1986). These remote signals hold the promise of enabling more accurate long range weather predictions to be made, as well as providing valuable clues to the workings of the global climate system.

Walker (1923; see also Walker and Bliss 1932), was the first to extensively study climatic anomalies associated with the SO. Walker established that the SO was associated with drought in India, and cool, wet winters over the southeastern United States. Berlage (1957) noted the association between drought in Indonesia and the SO. Nevertheless, it was not until the 1960s that the coupling between the SO and the ocean was demonstrated by Bjerknes (1966, 1969), who also established the connection between eastward shifts in equatorial convection over the tropical Pacific and a deepened wintertime Aleutian low over the North Pacific. This work set the stage for more detailed studies of the tropical and midlatitude climatic anomalies attributable to the SO (e.g., Wright 1977; Julian and Chervin 1978; Flohn and Fleer 1975; van Loon and Madden 1981; Horel and Wallace 1981; Rasmusson and Carpenter 1982; Ropelewski and Halpert 1986, 1987, 1989).

This study is an attempt to objectively identify those regions of the globe with significant temperature and precipitation anomalies associated with the SO. We are concerned here with both extremes of the oscillation: the phase associated with above normal sea surface temperature (SST) in the central and eastern equatorial Pacific (here termed warm events), and the opposite condition of below normal SST in the same region (cold events).

There have been many recent studies of the regional climatic effects of the SO. For example, Ropelewski and Halpert (1986) analyzed the effects of warm events on seasonal temperature and precipitation over North America, while Yarnal and Diaz (1986) identified climatic signals along the west coast of North America associated with both extremes of the oscillation. Other studies such as Rogers (1988) and Aceituno (1988) have focused on regional anomalies over Latin America during warm and cold events. However, few studies have been devoted to an overall, global view of climatic anomalies associated with the SO. Exceptions are Ropelewski and Halpert (1987, 1989) and Lau and Sheu (1988), who have identified areas with pronounced SO-related precipitation anomalies, and Glantz et al. (1987), who examined the extreme climatic anomalies worldwide during the 1982–83 event.

We have used station data to identify SO-related signals in temperature and precipitation over land, and

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ship data from the Comprehensive Ocean-Atmosphere Data Set (COADS; Woodruff et al. 1987) to examine SST signals over the world oceans. van Loon and Shea (1985, 1987), Meehl (1987), and Kiladis and van Loon (1988) have already examined the signals in sea level pressure (SLP), temperature, and precipitation over Australasia and the South Pacific Ocean, using similar techniques. Barnett (1984, 1985, 1988) has also examined the global SLP field during the evolution of an event in the SO. Since many of the signals discussed in the present paper have already been identified in previous regional work, we are concentrating here on giving a more comprehensive view of the global evolution of temperature and precipitation anomalies during both extremes of the SO. An effort is also made to provide references that the reader can refer to for further detail in specific areas of interest.

2. Data and methodology

The station data used in this study were obtained from the *World Monthly Surface Station Climatology* (also known as the *World Weather Records*), compiled at NCAR in the form of monthly mean surface pressure, temperature, and precipitation for several thousand stations around the globe. We have selected 1045 stations which provide the best long-period coverage possible for temperature and precipitation with this dataset. The distribution of these stations is shown in Fig. 1. The densely populated regions of Europe, Africa, southern Asia, and North America are well represented by these data. While there are large data gaps in certain areas such as central Asia, Africa, and equatorial South America, the problem is not too serious in most regions, since signals associated with the SO tend to be of broad spatial extent.

As seen from Fig. 1, the analysis will be primarily for continental regions, due to poor data coverage over the oceans. Since Kiladis and van Loon (1988) have already done a detailed analysis of the Pacific sector using similar techniques, that region will not be covered here.

The analysis approach uses all of the available data at each station to construct the mean anomaly in temperature and precipitation during extremes in the SO with respect to the entire station record. The entire station record was used as a base period to minimize the effects of temporal trends on the analysis. Possible temporal inhomogeneities in the data are neglected since they are not expected to significantly affect the results.

Cold and warm events were analyzed separately using a composite technique to examine the climatic signals associated with either extreme of the SO. Table 1 lists the year 0s of the events used, and is the same as that used in other recent studies of the SO (e.g., Bradley et al. 1987; Kiladis and van Loon 1988). These were chosen on the basis of a standard Southern Oscillation Index (SOI), which has been extended back to the year 1882 using recently discovered Tahiti pressure data (Ropelewski and Jones 1987), and an SST anomaly index for the eastern equatorial Pacific. The SST index represents the integrated seasonal SST anomaly within 4° of the equator from 160°W to the South American coast, based on COADS ship data. In order to qualify as an event, the SST anomaly had to be positive for at least three consecutive seasons, and be at least 0.5°C above the mean for at least one of these seasons, while the seasonal SOI had to remain below −1.0 for the same duration. The 1877 and 1880 warm events were also included in this list because of their apparent large magnitudes (Quinn et al. 1978; Kiladis and Diaz 1986).

Year 0 of a warm event is defined as in Rasmusson and Carpenter (1982) as the year when the SOI changes sign from positive to negative, and when central and

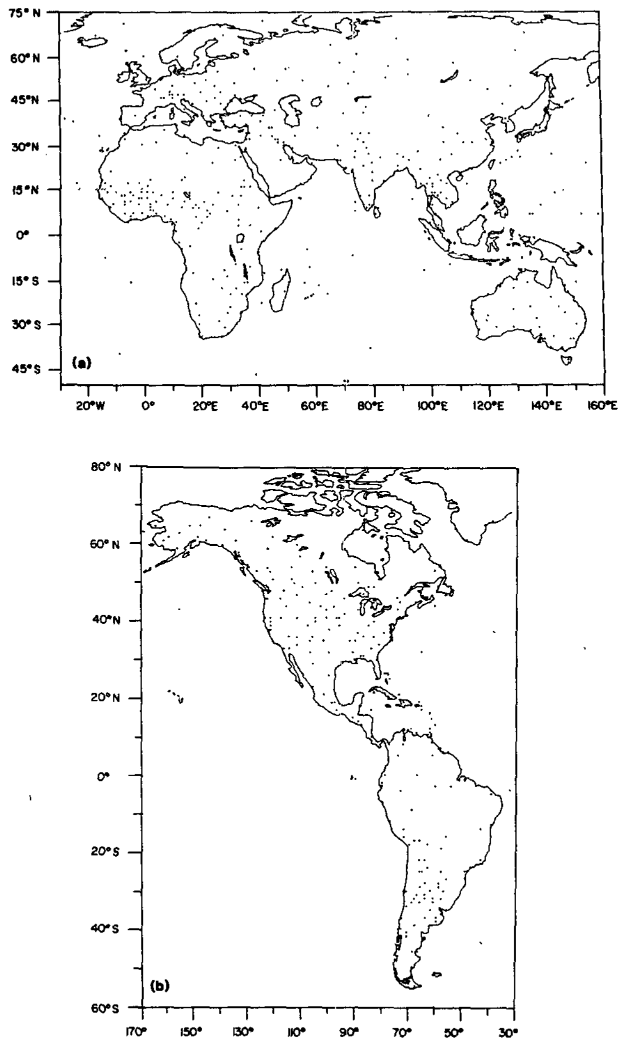

FIG. 1. Station distribution for (a) Eurasian sector, and (b) North American sector.

TABLE 1. List of year 0s of warm and cold events (1877–1988) used in the composite analysis.

Warm events (year 0)														
1877	1880	1884	1888	1891	1896	1899	1902	1904	1911	1913	1918	1923	1925	1930
1932	1939	1951	1953	1957	1963	1965	1969	1972	1976	1982	1986*			
Cold events (year 0)														
1886	1889	1892	1898	1903	1906	1908	1916	1920	1924	1928	1931	1938	1942	1949
1954	1964	1970	1973	1975	1988*									

* Recent events not used in the analysis.

eastern equatorial Pacific SST anomalies become strongly positive. Cold event year 0s are defined as having the opposite characteristics. Since we are interested in the sequence of development of events in the SO, we included only the first year of multiyear events and ignored successive years, regardless of their magnitudes.

The composite anomaly at each station was computed separately for the warm and cold events in Table 1 from the beginning of year -1 until the end of year $+1$, using the standard seasons December, January, and February (DJF), March, April, and May (MAM), etc. The t-values of the anomalies were also computed, as well as the t-value of the difference between the warm and cold event signals. A station had to have data for at least five events of each extreme of the SO to be included in the composite analysis. A signal was considered only if it was significant at the 95% level for a two-tailed test of the null hypothesis of no difference from the mean.

These t-values were calculated for temperature and precipitation at each station and then mapped for various stages of an event in the SO. Since in a spatial sense the data are heterogeneous, comprised of records of varying length, we make no attempt to analyze gradients in the anomalies. Instead, we are interested in whether an anomaly is significantly different from the long-term mean at spatially grouped stations, which gives an indication of whether we can expect the anomaly to occur in that region with any regularity during an event in the SO.

The COADS air temperature and SST data used are in the form of monthly means on a 4° grid. These data were analyzed in a fashion similar to that used for the station data, except here each 4° grid box was treated as a "station." One slight difference is that the anomalies were calculated with respect to the mean of three separate periods: 1854–1909, 1910–49, and 1950–79, which substantially reduces the effect of temporal trends in these data. Such trends are partially due to real changes in SST, and also to changes in observational techniques aboard ships over time (Ramage 1984; Woodruff et al. 1987).

In an initial analysis of both the land and ocean data, it was found that regions of large, statistically significant signals during warm events were characterized by equally large signals of the opposite sign during cold events. Essentially the same results were obtained by Meehl (1987), Bradley et al. (1987), Kiladis and van Loon (1988), and Ropelewski and Halpert (1989) in their analyses of temperature and precipitation during warm and cold events (see also Philander 1985). The implication is that the global climate system tends towards two approximately inverse states with respect to the SO: one corresponding to anomalously high and the other to anomalously low SST in the eastern equatorial Pacific; this is also supported by modeling studies (e.g. Lau 1985).

Given these initial results, we have based our spatial analysis on the difference between warm and cold events in order to test the hypothesis that the climatic anomalies at a given location differ between warm and cold events by a statistically significant amount, comprising two distinct populations. Our approach is to map the t-value of the composite warm minus cold event signals for each season. The sign of these mapped quantities represents the sign of the anomaly one would expect during a warm event, with the sign reversed for cold events. In order to better outline the regions possibly affected by the SO, we identify differences outside the middle 90% of the t-distribution, but only consider those regions having a core of signals significant at better than the 95% confidence level.

Since the magnitude of anomalies can vary greatly between events, one question arising in the use of this technique is whether a signal is dominated by a few large events. This is related to another question: How consistent is the signal at any given location from event to event? For the land data we have addressed this problem by simply calculating the percentage of consistent signals (PCS), defined as the percentage of events having anomalies of a sign consistent with the composite anomaly. For example, if a composite anomaly at a given station is positive during warm events and negative during cold events, the PCS gives the percentage of total (i.e., warm + cold) events that had a positive warm event anomaly and a negative cold event anomaly. The PCS is shown in table form for selected stations, along with the composite mean and anomaly for both warm and cold events. In ad-

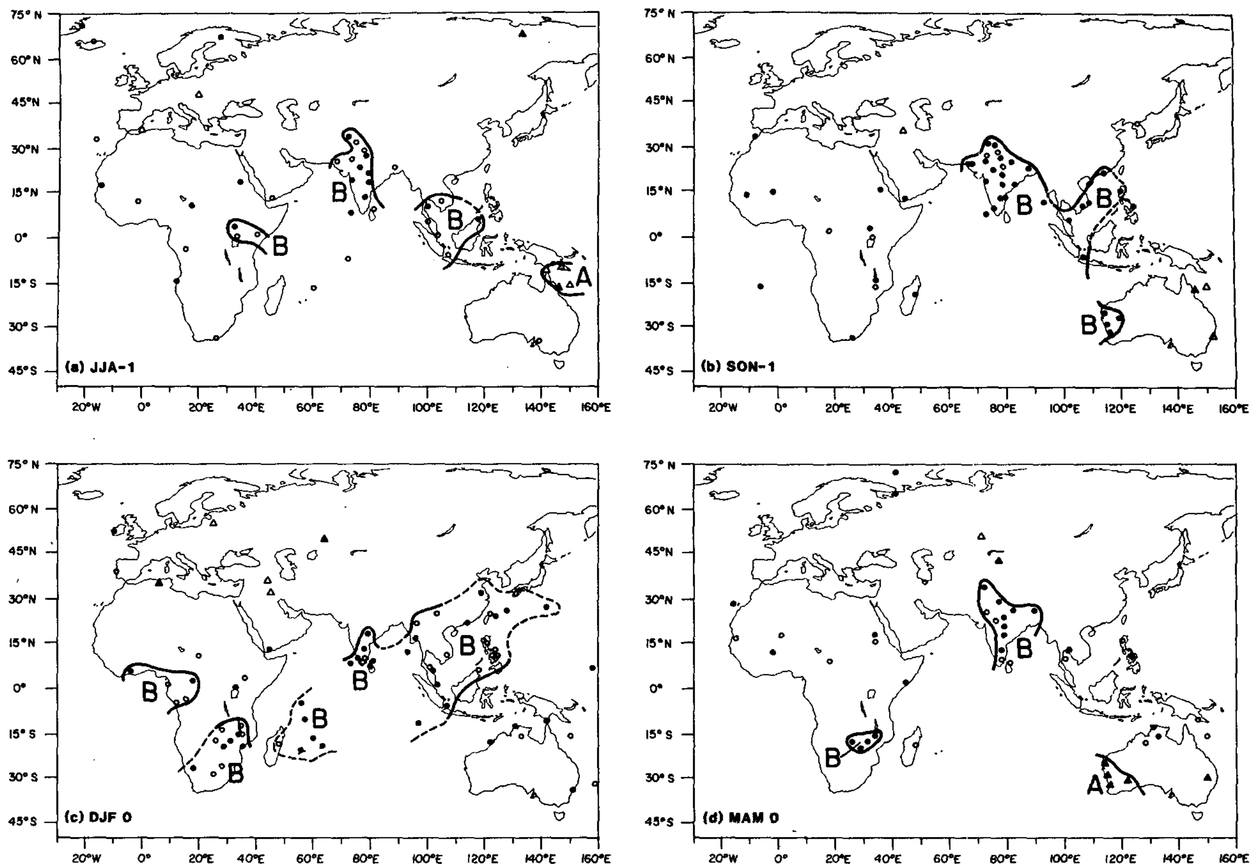

FIG. 2. Maps showing the sign of the composite difference (warm minus cold event) of Eurasian sector temperature for (a) JJA −1, (b) SON −1, (c) DJF 0, (d) MAM 0, (e) JJA 0, (f) SON 0, (g) DJF +1, and (h) MAM +1. Open circles (triangles) denote stations with negative (positive) differences significant at the 90% level. Signals significant at the 95% level are indicated by solid circles (triangles). The B represents below normal, and A above normal areas.

dition, we present in graphical form the total assemblage of normalized departures of temperature and precipitation during warm and cold events at some stations for seasons with strong signals. This illustrates the observed range of anomalies at a station during individual events, giving another measure of the reliability of the signal. As shown below, the opposition of signals between warm and cold events is usually very consistent for regions with statistically significant t-values.

As an additional measure of the chances of a given signal occurring at random, we have estimated the significance level of some of the strong signals revealed in the above analysis through the use of the hypergeometric distribution (Feller 1957). This method gives the probability that a minimum number of anomalies of one sign "K" will be chosen in "L" random trials from a sample consisting of "M" above and "N" below normal occurrences. Here $M + N$ is the total number of years available for a station, L the number of warm (or cold) events at the station, and K the number of years of one sign within the sample of events. This test, unlike the t-test, is very robust to skewness. The probabilities obtained from the hyper-geometric test are given as confidence level percentages in table form for selected stations.

For the ocean data we have indicated those grid points with a PCS of greater than 67%, i.e. where the sign of the anomaly was consistent with composite signal in at least two-thirds of the total events at that grid point. As in the land data, we only consider signals significant at the 95% level in the t-test.

It should be stressed that we are not necessarily searching for signals that are predictable, or that may themselves provide some basis for predictability of warm and cold events, since the inter-event variability of many of these anomalies is shown to be rather high. Instead we treat the signals as indicators of the state of the global atmosphere and ocean during the various phases of an event in the SO.

3. Results

Figures 2 through 5 are the composite temperature and precipitation difference maps (warm minus cold events) for the land areas analyzed in this study. Shown by open circles (triangles) are stations with negative (positive) differences significant at the 90% level. Sig-

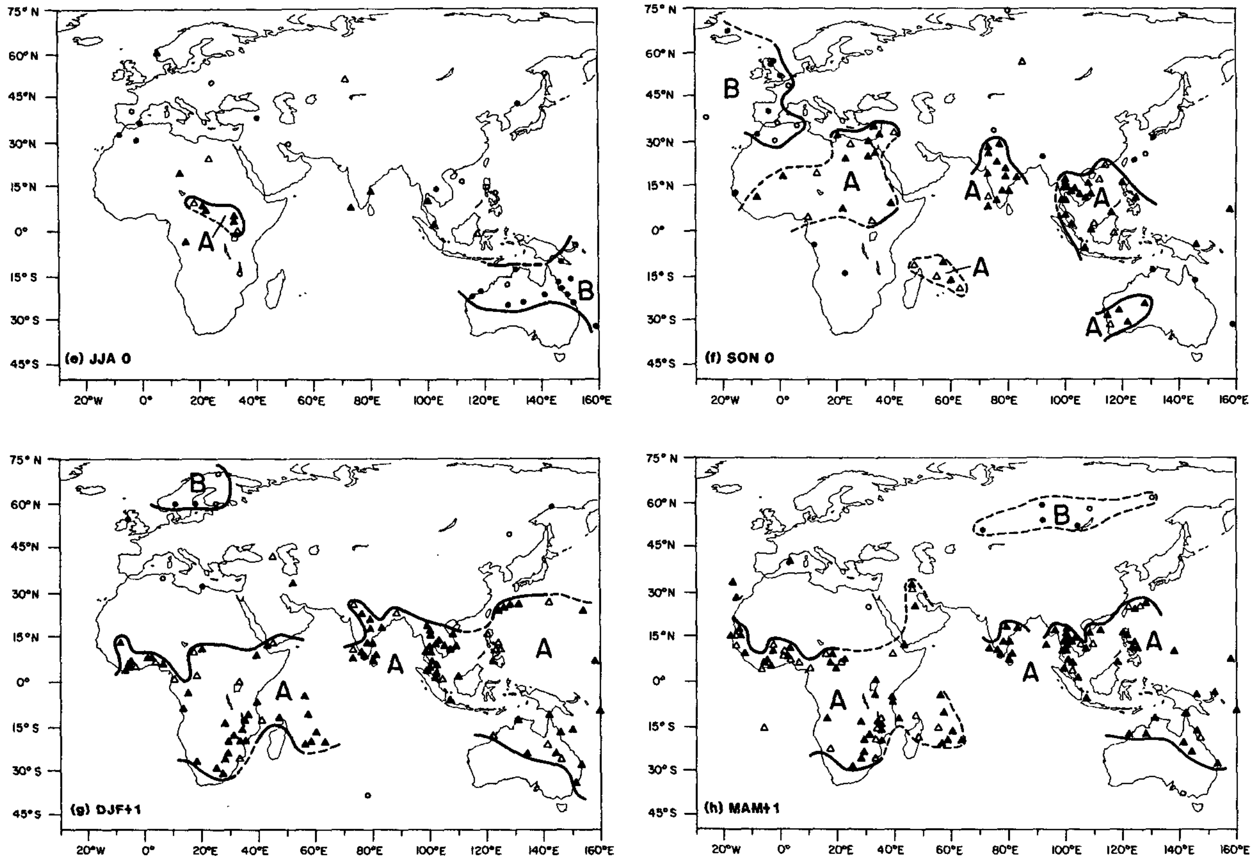

FIG. 2. (*Continued*)

nals significant at the 95% level are indicated by solid circles (triangles). Since the sign of the difference represents that of the warm event anomaly with respect to the mean, we will primarily refer to the signals one would expect during a warm event, bearing in mind that cold events would generally tend to have anomalies of the opposite sign.

On the maps we have subjectively outlined coherent regions which have a number of stations with significant signals. Where the data are too sparse to define a boundary, the outline is shown by a dashed line. For temperature, B (A) represents below (above) normal warm event conditions, with D (W) representing drier (wetter) than normal conditions on the precipitation maps. These quantities are mapped from June, July, and August of the year preceding the development of warm or cold event conditions in the equatorial Pacific (JJA −1), through March, April, and May of the year following the event development (MAM +1).

a. Eurasian sector

1) TEMPERATURE

The temperature signals over the Eurasian sector are shown in Fig. 2. During year −1 of a warm event we see a tendency for below normal temperatures in regions surrounding the Indian Ocean, with the strongest anomalies over India and southeasternmost Asia in JJA −1 and SON −1. There is also a tendency for below normal temperatures over Africa at this time, although the stations with significant signals are scattered. During DJF 0, the region of below normal temperatures expands to include southeastern China and the Philippines, with strong signals at Indian Ocean stations, southern Africa, and the west coast of equatorial Africa. This pattern weakens into MAM 0, and begins to reverse sign by JJA 0. Table 2 shows the composite mean DJF 0 temperatures at Trincomalee, Sri Lanka during warm and cold events. Due to the generally low variance in tropical temperature, there is only a 0.5°C difference between warm and cold event mean temperature. Nevertheless, temperatures were below the mean in 84% of the 25 warm events, and above the mean in 65% of the 20 cold events at Trincomalee, and in the case of warm events this recurrence is significant at better than the 99.9% level based on the hypergeometric distribution. Figure 6a plots the normalized anomaly of Trincomalee temperature during DJF 0 for all warm and cold events with data. With a few exceptions, temperatures during warm and cold events are well separated into two distinct populations in this season at Trincomalee.

TABLE 2. Composite warm and cold event seasonal mean temperature and departures from the mean (in °C) at stations with strong SO signals. Also shown is the number of events (N), the significance level (Sig) based on the hypergeometric test, the percentage of events consistent with the composite (PCS), and the combined warm and cold event PCS (total PCS) for each station (see text for an explanation of the PCS).

		Warm event					Cold event					
Station	Season	Mean	Anom	PCS	N	Sig	Mean	Anom	PCS	N	Sig	Total PCS
					Eurasian sector							
Trincomalee 8.6°N, 81.2°E	DJF 0	25.9	−0.2	84	25	99.9	26.4	+0.3	65	20	91.7	76
Darwin 12.5°N, 130.9°E	JJA 0	25.2	−0.4	71	24	98.1	26.1	+0.5	75	20	99.2	73
Madrid 40.4°N, 3.7°W	SON 0	13.7	−0.3	65	26	95.6	14.4	+0.4	74	19	98.2	69
Naha 26.2°N, 127.7°E	DJF +1	17.0	+0.3	68	22	96.7	16.4	−0.3	71	17	94.5	69
Colombo	DJF +1	26.7	+0.3	77	26	99.6	26.2	−0.2	79	19	99.3	78
6.9°N, 79.9°E	MAM +1	28.1	+0.3	73	26	99.2	27.6	−0.2	74	19	97.6	73
Bulawayo	DJF +1	22.1	+0.5	65	20	96.8	21.2	−0.4	82	17	99.1	73
20.2°S, 28.6°E	MAM +1	19.0	+0.5	76	21	99.9	18.1	−0.4	76	17	99.0	76
					American sector							
Rio de Janeiro 22.9°S, 43.2°W	JJA −1	20.5	−0.3	77	26	99.9	21.0	+0.2	65	20	82.4	72
Juneau	JJA −1	12.6	−0.2	77	18	99.7	13.2	+0.4	56	16	58.7	67
58.3°N, 134.4°W	DJF +1	−0.8	+1.3	74	19	97.0	−3.3	−1.2	64	14	88.5	70
Prince Albert 53.2°N, 105.7°W	DJF 0	−18.2	−0.9	70	20	96.5	−15.5	+1.8	72	18	97.3	71
Goya 29.2°S, 59.3°W	JJA 0	15.1	+0.4	65	20	83.2	13.8	−0.9	80	15	99.4	71
Manaus	DJF +1	26.8	+0.6	94	17	99.9	25.8	−0.4	100	13	99.9	97
3.1°S, 60.0°W	MAM +1	26.4	+0.3	76	17	99.4	25.7	−0.4	100	13	99.9	86
Galveston	DJF +1	12.2	−0.9	84	25	99.9	14.1	+1.0	85	20	99.9	84
29.1°N, 94.9°W	MAM +1	19.9	−0.4	72	25	99.3	20.7	+0.4	75	20	99.4	73
Jacksonville 30.3°N, 81.7°W	DJF +1	12.7	−0.8	73	26	99.7	14.5	+1.0	75	20	96.7	74
New Orleans 30.0°N, 90.1°W	DJF +1	12.2	−1.0	88	26	99.9	14.2	+1.0	75	20	98.6	82
Winnipeg	DJF +1	−14.9	+1.6	77	26	99.8	−17.8	−1.8	70	20	98.8	74
49.9°N, 97.1°W	MAM +1	2.7	+0.7	65	26	82.1	0.8	−1.2	65	20	96.7	65
Edmonton	DJF +1	−9.8	+2.1	83	24	99.9	−13.7	−1.8	75	20	99.0	79
53.6°N, 113.5°W	MAM +1	4.0	+0.5	54	24	57.0	2.4	−1.1	70	20	95.3	61
Abilene 32.4°N, 99.7°W	MAM +1	17.2	−0.8	83	23	99.9	18.4	+0.4	60	20	74.1	72

During JJA 0 a large area of negative temperature anomalies covers northern Australia, with significant positive anomalies occurring in central Africa just north of the equator. The high consistency of this signal at Darwin is illustrated in Table 2 and Fig. 6b.

In SON 0 temperature anomalies are now opposite to what they were in the previous year, with above normal temperatures strengthening and expanding to cover almost the entire tropical portion of the sector during DJF +1 and MAM +1. The tropical temperature signal during DJF +1 and especially MAM +1 is very consistent, exhibiting a good separation between warm and cold events at such widely scattered locations as Naha, on Okinawa Island; Colombo, Sri Lanka; and Bulawayo, Zimbabwe (Table 2; also see Fig. 6c for Bulawayo).

The tendency for tropical tropospheric temperature anomalies to follow the anomaly of SST in the equatorial Pacific was first demonstrated by Newell and Weare (1976), and verified in subsequent studies (e.g., Angell 1981; Bradley et al. 1987). It is well known that eastern equatorial Pacific SST is, on average, below normal during year −1 (Rasmusson and Carpenter 1982; see section 3c), and the year −1 signal of below normal tropical temperatures reflects this. In fact, the general tendency for opposition between year −1 and year 0 signals in temperature, precipitation, and sea level pressure over the Pacific and Indian sectors has been demonstrated in other recent studies (e.g., Meehl 1987; van Loon and Shea 1985; Kiladis and van Loon 1988), and will be shown below to be present in other regions of the globe as well.

Three other interesting anomalies are the below normal temperatures over western Europe in SON 0, over Scandinavia during DJF +1, and in north-central Asia in MAM +1. The Scandinavian and Asian signals are not consistent among events, but Madrid's signal is fairly reliable during SON 0 (Table 2).

2) PRECIPITATION

Warm events are known to be associated with large precipitation departures, especially in the tropics, but so far most work on this topic has focused on anomalies which develop once the event is well underway. Less is known about precipitation anomalies which precede the development of warm events, or those associated with cold events. Maps of warm minus cold event precipitation anomalies over Eurasia during year −1 (Fig. 3) show enhanced rainfall over the Indonesia/northern Australia region during JJA −1. This signal is especially striking at Ambon, Indonesia, which records an average difference of 290 mm of precipitation/month between warm and cold events (Table 3 and Fig. 7a), although over Indonesia as a whole this anomaly occurs during the dry season, lessening its economic consequences. Nevertheless, the signal strengthens in SON −1, giving Indonesia a tendency for heavier than normal rainfall at the beginning of the wet season preceding warm events. There are a few other areas of scattered anomalies during SON −1, such as a moist anomaly just north and west of the Black Sea and a dry anomaly in eastern equatorial Africa, although these areas are small and their signals variable between events (data not shown).

In DJF 0, a well-defined region of wetter than normal conditions affects southeastern Africa, strengthening into MAM 0. This signal, especially the MAM 0 wet conditions during warm events, is particularly strong at Harare, Zimbabwe (Table 3 and Fig. 7c). These anomalies occur at the end of the wet season in this region, resulting in an extension of the rainy season during warm events and an early termination during cold events. Note also the appearance of a wet anomaly in West Africa in DJF 0, which expands eastward into MAM 0. This signal is characterized, however, by rather large inter-event variability (not shown).

By JJA 0 the equatorial Pacific SST warming is, on average, well under way (e.g., Rasmusson and Carpenter 1982), with a large area of negative precipitation anomalies beginning to affect southern Asia in Fig. 3e. Most notable is the weak monsoon precipitation over India, which has been discussed in detail in several other studies (e.g., Shukla and Paolino 1983; Rasmusson and Carpenter 1983; Parthasarathy and Pant 1985). The fact that cold events have a clear tendency toward excessive precipitation over India is displayed in Table 3, which shows that the signal is more pronounced at stations to the south (Bangalore, Madras) than for stations in northern India (e.g., New Delhi, Darjeeling). The marked separation between warm and cold event precipitation at Bangalore is illustrated in Fig. 7e. Somewhat weaker anomalies extend eastward and southward along the Malay Peninsula, to affect parts of Indonesia and eastern Australia, while rainfall is enhanced from the Philippines eastward into the intertropical convergence zone (ITCZ) of the western tropical Pacific (see Meehl 1987; Kiladis and van Loon 1988). In Africa, there is a weak tendency for drought at some stations in the Sahel and in Ethiopia, and wetter than normal conditions in the western Mediterranean.

Rainfall anomalies in the Australasian region are most intense in SON 0, with drier than normal conditions occuring over the Philippines, Indonesia, New Guinea, and most of Australia in Fig. 3f. Once again the anomalies are especially pronounced at Ambon, Indonesia, where cold event precipitation averages more than 250% than during warm events (Table 3 and Fig. 7b). A comparison of Fig. 7b with Fig. 7a for JJA −1 reveals a striking reversal in precipitation anomalies at Ambon between year −1 and year 0. In Australia, the strongest anomalies appear in the southeast (e.g., Adelaide, Table 3). Dry anomalies also show up in parts of central India and along the west coast of Africa. The wetter than normal conditions over southeastern India and Sri Lanka at this time are well documented (e.g., Rasmusson and Carpenter 1983; Ropelewski and Halpert 1987); this signal is fairly reliable, as for example at Trincomalee (Table 3).

The weak dry anomaly along the east coast of Africa in the previous year now has the opposite sign. The signal is mixed at these stations, but appears to be much more stable over the Indian Ocean, as at Mahe, in the Seychelle Islands (Table 3). Lack of adequate data over the central Indian Ocean precludes documentation of the probable large signal there. The signal along the coast of West Africa shows up well at Dakar, Senegal (Table 3), especially for cold events. The dry signal along the coast farther south is, however, less reliable.

An interesting wet anomaly extends from Iraq into the Soviet Union during SON 0. Although the extent of this region is not very well defined due the lack of data, we see that at Tashkent (Table 3) the signal is very significant, particularly for warm events. Farther west, the tendency for above normal warm event precipitation during SON 0 and below normal during cold events at Madrid is also clearly evident in Table 3.

Eurasian precipitation anomaly patterns have many similarities during DJF +1 and MAM +1, with dry conditions over northern Australia and in the ITCZ of the western Pacific, and wet conditions at some stations in eastern China and southern Japan. Of particular interest are the dry anomalies in southern Africa, which occur during the rainy season there, and are opposite to the strong wet anomalies of the previous year in Figs. 3c and 3d. At Kimberly, South Africa, warm and cold events show quite significant differences in precipitation during both seasons, with large and highly significant anomalies at several other stations farther north as well, as at Harare, Zimbabwe during MAM +1 (Table 3). The marked tendency for a biennial reversal in the precipitation anomalies at Harare can be seen by comparing Fig. 7d for MAM +1 with Fig. 7c for MAM 0. Nicholson and Entekhabi (1986) dem-

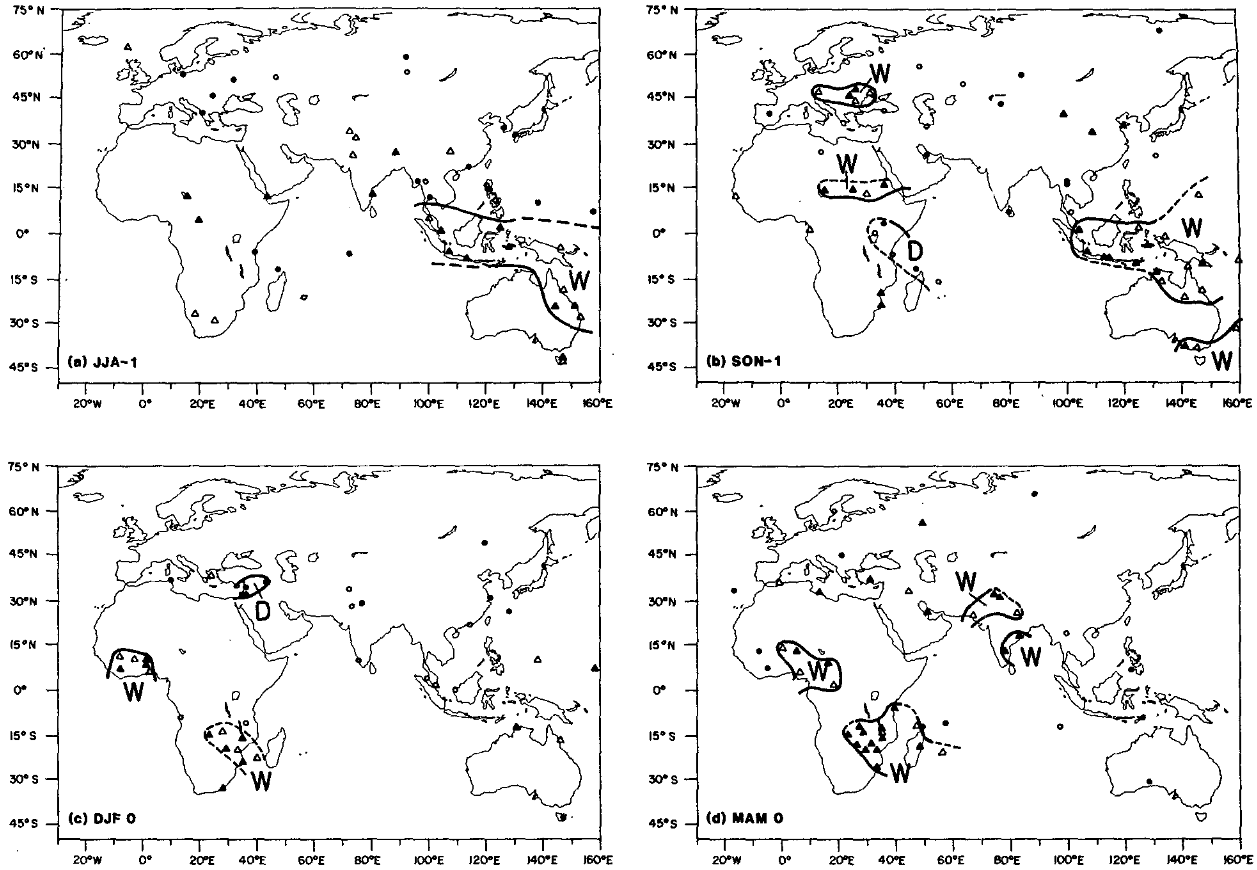

FIG. 3. As in Fig. 2, except for Eurasian sector precipitation. The W represents wetter than normal, and D drier than normal conditions.

onstrated that the strongest coherence between the SO and African rainfall was in the 2.2 to 2.4 year range, presumably reflecting the biennial tendency of the SO.

Another notable anomaly during DJF +1 and MAM +1 is the tendency for wetter than normal conditions during warm events over parts of Europe. Although most of the stations exhibit substantial overlap between warm and cold event anomalies, the signal is rather more reliable at some stations such as Zurich, Switzerland, during DJF +1 (Table 3). During MAM +1 this signal extends farther eastward into the western Soviet Union, as for example, at Kiev (Table 3).

b. *American sector*

1) TEMPERATURE

Tropical temperatures in the American sector show primarily the same warm event signal as in the Eurasian sector, with below normal temperatures during year −1 and above normal during year 0 (Fig. 4). If anything, the strength of these anomalies is even greater than those in the Eurasian sector, perhaps because of the proximity of this region to the anomalous conditions of the eastern equatorial Pacific during extremes of the SO. In addition, the midlatitude anomalies in the American sector are more pronounced than over Eurasia during warm and cold events, since teleconnections from the tropical Pacific more directly affect the extratropical circulation over the North and South Pacific oceans.

Already during JJA −1 large areas of the tropical and subtropical Americas are experiencing below normal temperatures during warm events (Fig. 4a) with the opposite tendency during cold events, as shown at Rio de Janeiro (Table 2). Three other notable anomalies are the above normal warm event temperatures in Patagonia and Arctic Canada, and the below normal temperatures in northwestern North America. Of these regions, the latter shows the most reliable signal, especially for warm events (e.g., Juneau, Table 2).

The area affected by above normal temperatures in southeastern South America expands into SON −1. Note that this anomaly also affects the South Atlantic to the east of this region, as in the Falkland Islands and South Georgia. This signal disappears in DJF 0, and although the tropical signal persists, it is less consistent than in previous seasons (data not shown). What is more striking during this season is the vast area affected by below normal warm event temperatures in central Canada and northwestern North America. The strength

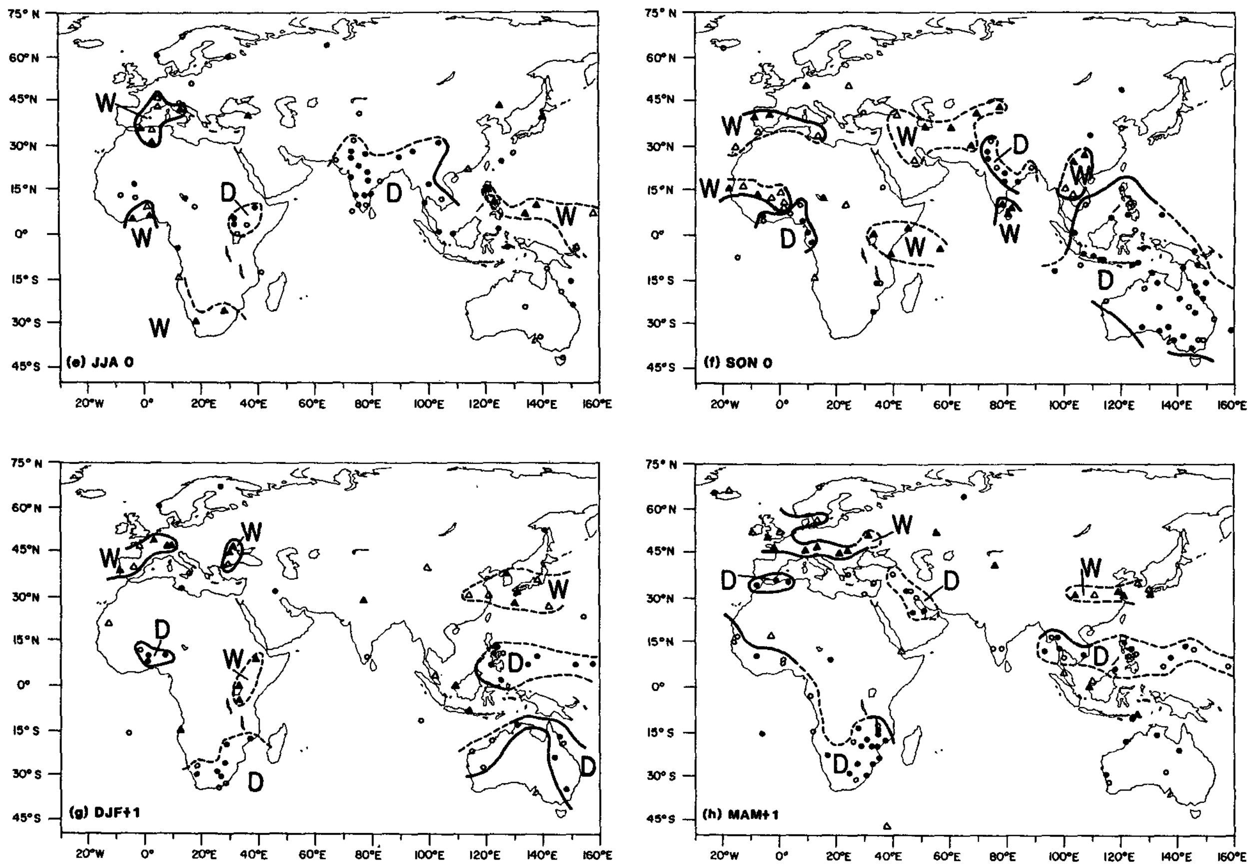

FIG. 3. (*Continued*)

and consistency of this signal is especially apparent in Canada at Prince Albert (Table 2), although at most other stations there is more overlap.

During MAM 0, when on average, SST anomalies go from negative to positive along the west coast of South America, coastal air temperature does likewise. This anomaly expands into the subtropics of South America during JJA 0 (Fig. 4e), giving a pattern which is now opposite to that of the previous year (compare Fig. 4a). As in year −1, this signal is quite consistent in subtropical South America (e.g., Goya, Argentina, Table 2).

Although the area covered by below normal temperature in northern North America during SON 0 is extensive (Fig. 4f), this signal is not very reliable (data not shown). Above normal temperatures now begin to affect Central America and the Caribbean, with the large anomalies in south-central South America weakening considerably compared to JJA 0 (compare Fig. 4e).

As in the Eurasian sector, the areal extent and strength of the tropical American temperature anomalies are greatest in DJF +1 and MAM +1. There is very little overlap in the temperature anomalies between warm and cold events, not only along the west coast of South America, which would be directly affected by SST anomalies in the El Niño region, but also at stations on the other side of the Andes, such as Manaus, Brazil (Table 2 and Fig. 6d).

Extensive areas of North America are also affected by strong temperature anomalies during DJF +1 and MAM +1. Below normal temperatures in the southeastern United States during warm events have been well documented since Walker's time (e.g., Ropelewski and Halpert 1986). This signal is very consistent at Galveston, New Orleans, and Jacksonville (Table 2) during both warm and cold events. Figure 6e illustrates the high reliability of this signal at New Orleans. While the signal is maximized in DJF +1, it is still strong at Galveston in MAM +1, although less so in the southern Great Plains at this time, as at Abilene, Texas (Table 2).

The other pronounced temperature signal in North America is the very strong tendency for above normal values in Alaska, southern Canada, and along the west coast. Walker (1932) first discovered this signal in western North America, and it has been linked to the so-called Pacific–North American (PNA) circulation pattern by Horel and Wallace (1981). The Canadian portion of this pattern has been further substantiated

TABLE 3. As in Table 2, except for seasonal mean precipitation. The precipitation value is given in millimeters and represents a monthly mean for that season.

		Warm event					Cold event					
Station	Season	Mean	Anom	PCS	N	Sig	Mean	Anom	PCS	N	Sig	Total PCS
					Eurasian sector							
Ambon	JJA −1	662	+146	71	21	99.9	372	−144	88	17	99.7	79
3.7°S, 128.2°E	SON 0	95	−71	91	22	99.9	243	+77	81	16	99.9	87
Harare	MAM 0	65	+14	70	20	99.2	32	−19	94	17	99.9	81
17.9°S, 31.0°E	MAM +1	38	−13	80	20	98.8	66	+15	82	17	99.9	81
Madras 13.1°N, 80.3°E	JJA 0	76	−16	77	26	99.8	106	+14	65	20	93.3	72
Bangalore 13.0°N, 77.6°E	JJA 0	86	−22	88	26	99.9	131	+23	70	20	99.8	80
New Delhi 28.7°N, 77.3°E	JJA 0	144	−13	58	26	71.4	164	+7	55	20	72.4	57
Darjeeling 27.1°N, 88.3°E	JJA 0	624	−20	62	21	44.1	661	+17	53	17	87.3	58
Adelaide 35.0°S, 138.5°E	SON 0	36	−5	69	26	94.5	47	+6	65	20	94.6	67
Trincomalee 8.6°N, 81.2°E	SON 0	260	+32	72	25	99.6	198	−30	80	20	99.4	76
Mahe 4.6°S, 55.5°E	SON 0	219	+47	88	16	99.9	130	−42	85	13	98.9	87
Dakar 14.7°N, 17.5°W	SON 0	76	+11	53	19	89.4	52	−13	88	17	99.1	70
Tashkent 41.3°N, 69.3°E	SON 0	31	+8	64	25	99.7	20	−3	80	20	96.8	71
Madrid 40.4°N, 3.7°W	SON 0	58	+13	73	26	95.6	36	−7	68	19	98.1	71
Kimberly	DJF +1	49	−11	65	26	94.3	74	+14	65	20	89.8	65
28.7°S, 24.8°E	MAM +1	37	−8	62	26	75.3	53	+8	60	20	77.6	61
Zurich 47.4°N, 8.4°E	DJF +1	73	+7	62	26	84.6	54	−12	80	20	99.7	70
Kiev 50.5°N, 30.5°W	MAM +1	55	+9	64	25	97.3	39	−7	75	20	96.7	69
					American sector							
San Jose	JJA −1	284	+40	71	24	99.5	208	−36	79	19	99.2	75
9.9°N, 84.1°W	JJA 0	206	−38	75	24	98.7	315	+71	80	20	99.9	77
Salta 24.8°S, 65.5°W	SON −1	39	+8	53	19	88.5	18	−13	94	16	99.8	72
Santiago 27.9°S, 64.4°W	JJA 0	92	+22	62	26	99.4	51	−19	85	20	98.1	72
St. Clair 10.7°N, 61.5°W	DJF +1	59	−19	85	20	99.9	92	+14	69	16	96.1	78
Bogota 4.6°N, 74.1°W	DJF +1	46	−16	79	24	99.9	75	+13	74	19	96.1	77
Abilene 32.4°N, 99.7°W	DJF +1	35	+8	70	23	99.6	18	−9	95	20	99.9	82
Nassau 25.1°N, 77.4°W	DJF +1	57	+15	64	25	99.9	28	−14	89	18	97.4	74

by Ropelewski and Halpert (1986). In Figs. 4g and 4h positive differences extend all the way to the east coast in Newfoundland. The most impressive signal is seen at stations such as Edmonton and Winnipeg, where warm event winters average greater than 3°C warmer than cold event winters (Table 2). With few exceptions, the separation of winters is quite spectacular at Winnipeg in Fig. 6f. Along the west coast the anomalies are almost as pronounced, as for example, at Juneau (Table 2). At Edmonton and Winnipeg the signal still persists, but is much weaker, during MAM +1 (Table 2).

2) PRECIPITATION

Precipitation anomalies over the American sector during warm and cold events are generally not as pronounced as over the Eurasian sector. During JJA −1 there is a tendency for wet conditions in the Caribbean region (Fig. 5a), as at San Jose, Panama (Table 3), along with dry conditions over scattered stations in the western United States and southern South America. Although the consistency of the latter signals is not great (data not shown), they are generally opposite in sign to those found a year later (Fig. 5e). The same is true for signals during SON −1, with wet conditions in western Argentina (e.g., Salta, Table 3) and dry conditions farther south and in the southwestern United States.

In DJF 0 there is a relatively strong wet signal over western Canada, with weaker signals of dryness over the south coast of Alaska and the southeastern United States. The small region of above normal precipitation in northern South America is not very reliable. There are no strong signals during MAM 0 in our analysis, although in this season heavy rains are sometimes observed in coastal regions of northern Peru and Ecuador during the onset of El Niño conditions offshore (e.g., Deser and Wallace 1987).

Composite precipitation anomalies become more pronounced over the Americas during JJA 0. The drier than normal conditions around the Caribbean basin are distinctly inverse to those of the previous year at stations such as San Jose, Panama (Table 3). The wet winter conditions in southern South America are highly significant at Santiago, Chile (Table 3), and are once again opposite to the JJA −1 signal. A few stations in western North America also reverse the sign of their precipitation anomaly compared to JJA −1, but their signal is weak and unreliable.

The dry/wet dipole over northern/southern South America continues into SON 0 and DJF +1, with well-defined signals at St. Clair, Trinidad and Bogota, Colombia (Table 3). Unresolved by our station distribution is the wet signal along coastal Peru and the tendency for the altiplano of southern Peru to share the dry signal of the interior during these seasons (e.g., Quinn and Neal 1983).

An interesting, but weak, anomaly occurs during SON 0 in the southwestern United States in Fig. 5f, with an implied tendency for an early beginning to the rainy season during warm events and a delayed beginning during cold events. Again, this signal is opposite to that of the previous year (Fig. 5b), strengthening the case for its relationship to the SO.

Another well-known warm event precipitation anomaly occurs during winter along the Gulf Coast of the United States, which has a tendency for wet conditions along with the previously discussed below normal temperatures observed at this time (e.g., Douglas and Englehart 1981; Ropelewski and Halpert 1986). Figure 5g indicates that this signal also affects inland Texas, northern Mexico, Cuba, and the Bahamas as well (see Rogers 1988), and is a very reliable signal at these locations (e.g., Nassau, Table 3; Abilene, Fig. 7f). This anomaly spreads into the High Plains area of the United States during northern spring (Fig. 5h).

c. Global SST

We show in Fig. 8 maps of the composite warm minus cold event SST anomaly from JJA −1 through MAM +1. In these figures the sign of the anomaly is shown at those grid boxes where the SST difference is statistically different from zero at greater than the 90% level, *and* which have a PCS of better than 67%. Positive differences significant at the 95% (90%) level are depicted as heavy (light) plus signs, and negative differences by heavy (light) minus signs. From the figures it can be seen that after application of the PCS criterion the vast majority of the boxes are significant at better than the 95% level. For the most part, the anomaly patterns agree with those found in previous studies (e.g., Pan and Oort 1983; Wright et al. 1985). Air temperature anomalies (not shown) match very closely those of SST.

Figs. 8a and 8b reveal the strong tendency for below normal SST in the eastern tropical Pacific during JJA −1 and SON −1 of a warm event, as was shown by Rasmusson and Carpenter (1982). Significant negative anomalies also occur along the northwest coast of North America, the east coast of South America, the southwest coast of Africa, and in the central and northern Indian Ocean. A region of below normal SST also develops during SON −1 in the tropics of the North Atlantic and in the South China Sea. Positive anomalies are found around southernmost South America, Indonesia, and in the region of the South Pacific convergence zone (SPCZ). For the most part, the sign of these SST anomalies matches that of air temperature along adjacent coastal areas in Figs. 2 and 4. Positive SST anomalies in the SPCZ region appear to be a response to enhanced northerly winds over this region at this time (van Loon and Shea 1987), while the Indonesian anomalies appear to be a result of the weaker than normal southeasterly basic flow, reducing oceanic mixing and evaporation (Hackert and Hastenrath 1986). These circulation anomalies arise from the anomalously low SLP centered over Australia at this stage of the event (van Loon and Shea 1987; Kiladis and van Loon 1988). There is good evidence that these SST anomalies play a key role in initiating the transition from one state of the SO to the other (see Nicholls 1984a,b; Hackert and Hastenrath 1986; van Loon and Shea 1987; Meehl 1987; and Storch et al. 1988).

By DJF 0 the negative anomalies in the eastern equatorial Pacific are weakening, although the area covered by negative departures is still increasing over the Indian Ocean and South China Sea. Coincident with this, there has been a rapid transition to below normal SST in the Indonesian region. At least some of this signal could be explained by a higher than normal frequency of cold air outbreaks out of Asia at this time (Kiladis and van Loon 1988), along with anomalously high wind stress and cloudiness over the southern monsoon region observed at this stage of an event

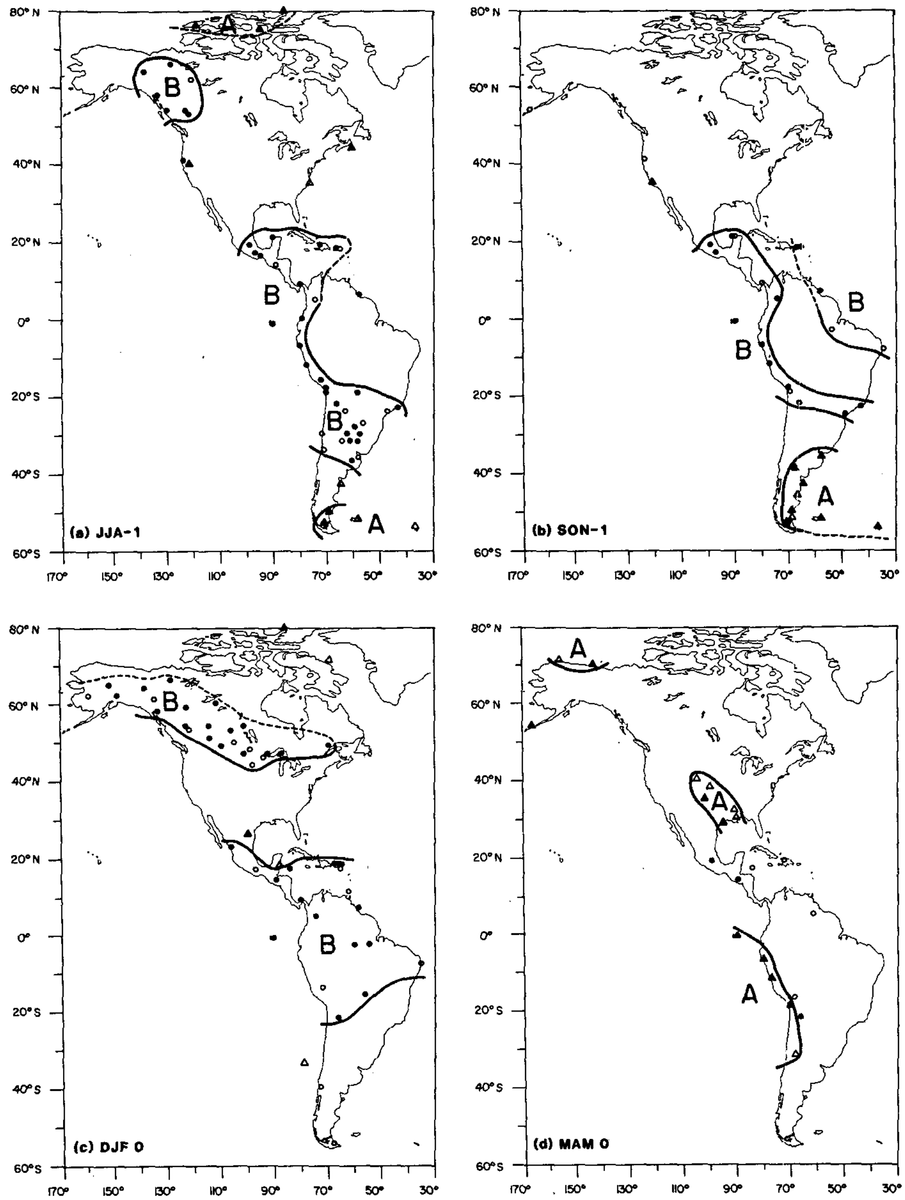

FIG. 4. As in Fig. 2, except for American sector temperature.

(Nicholls 1984a; Hackert and Hastenrath 1986). Figure 9a is a map of subjectively contoured composite SST differences between warm and cold events during DJF 0. Outside of the tropical Pacific, the absolute value of most of the differences are less than 1°C, with the largest differences located near the equator just to the east of the date line.

On average, above normal SST begins to appear in the eastern equatorial Pacific in MAM 0 (Fig. 8d), in agreement with Rasmusson and Carpenter's (1982) composite analysis using a smaller sample. As with the land data, overall anomalies are weakest at this time, indicative of a transition between cold and warm event conditions.

The area covered by positive anomalies has increased rapidly in the eastern Pacific by JJA 0 (Fig. 8e), and extends southward along the South American coast to about 40°S. Negative anomalies persist around Southeast Asia, Indonesia and in the Coral Sea, and begin to appear at this time in the region of the SPCZ. This anomaly strengthens substantially in SON 0 (Fig. 8f), which is also the time of maximum negative anomaly in the waters surrounding Indonesia. As shown in Fig. 3f, a strong signal of below normal precipitation occurs

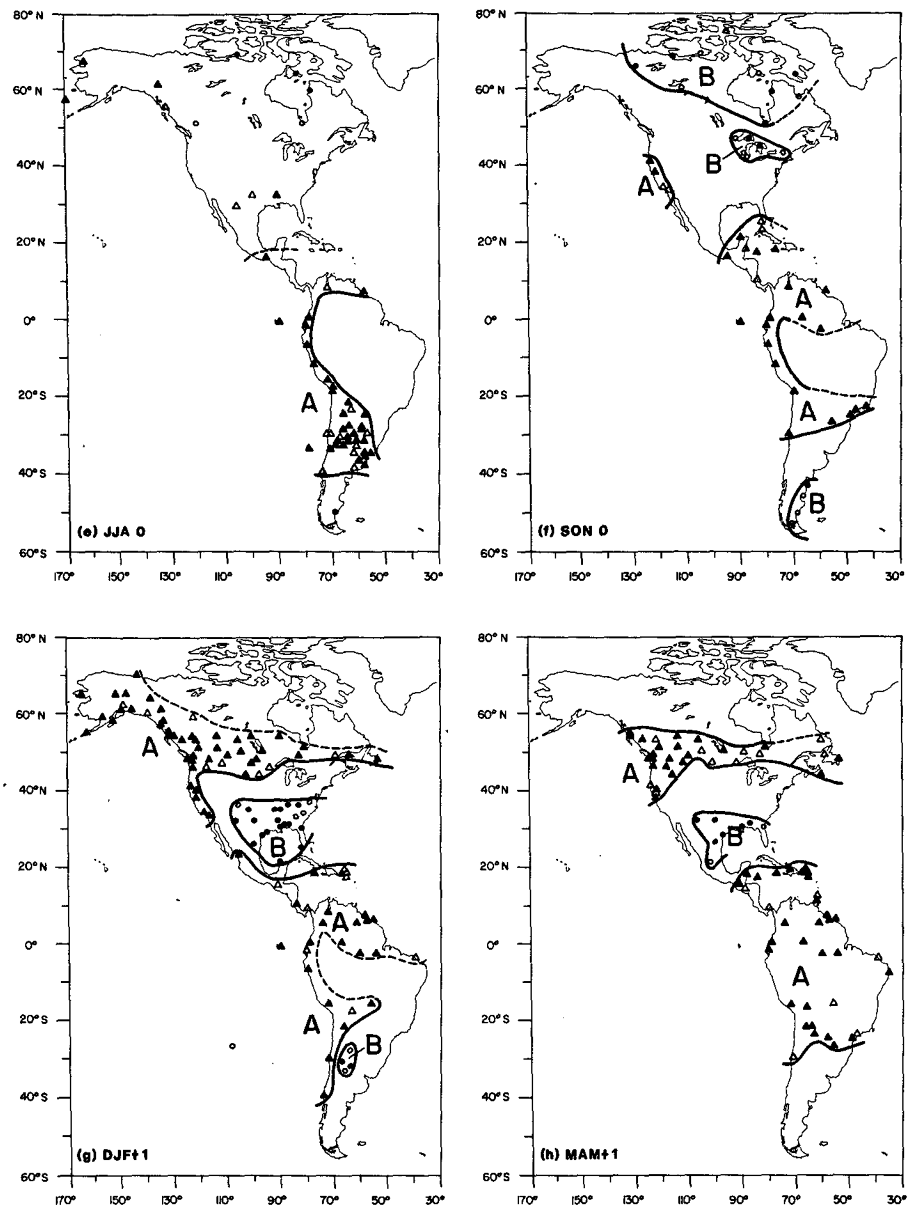

FIG. 4. (*Continued*)

in these regions in this season (see also McBride and Nicholls 1983; Kiladis and van Loon 1988). The anomalies in the eastern South Atlantic, central and northern Indian Ocean, and tropical North Atlantic, which were in phase with the eastern Pacific and negative during year −1, now begin to become positive. Positive anomalies also extend poleward along the Pacific coast of North and South America, in response to Kelvin wave propagation along the eastern boundary of the Pacific basin (Enfield and Allen 1980). Note the development of negative SST anomalies in the central North Pacific, which appear to be tied to an increase in surface midlatitude westerlies (e.g., Bjerknes 1966). This same forcing likely operates in the subtropical South Pacific, where the negative SST and surface westerly wind anomalies extend from the SPCZ well into midlatitudes (van Loon 1984; van Loon and Shea 1986; Meehl 1987).

The pattern in DJF +1 is similar to that in SON 0 (Fig. 8g), except that anomalies have weakened in the SPCZ and reversed sign and become positive around Indonesia. In most other areas this is the time of maximum anomaly, and Fig. 9b shows that the differences approach 3°C in the eastern equatorial Pacific. Positive

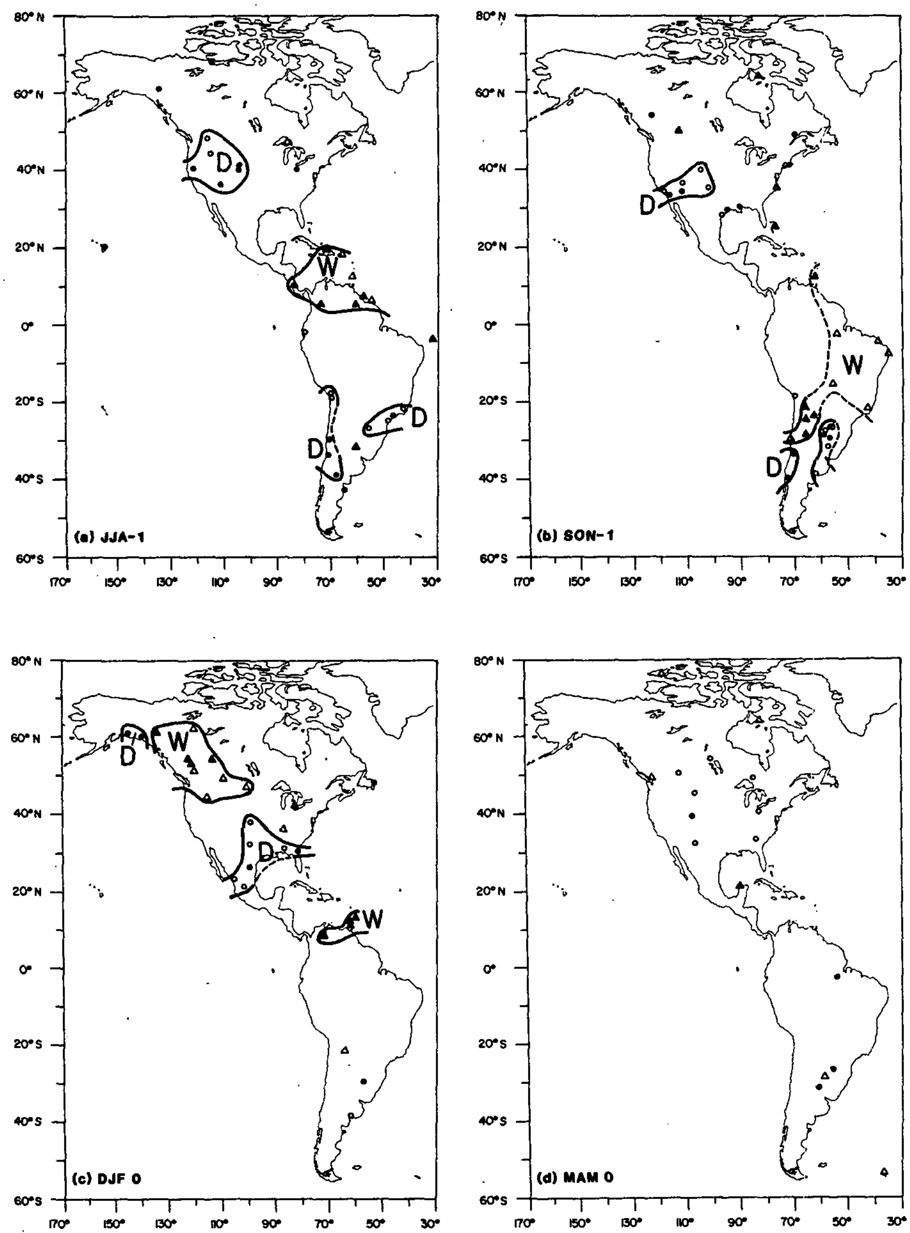

FIG. 5. As in Fig. 3, except for American sector precipitation.

differences now exceed 0.5°C in a large area of the Indian Ocean. The extension of the positive difference along the west coast of North America, all the way into the Gulf of Alaska, is very pronounced. It appears that at least part of this signal is a response to changes in the wind-driven oceanic circulation over the North Pacific (e.g., Emery and Hamilton 1985).

In MAM +1 the anomalies in the eastern Pacific are beginning to break down, although the in-phase signal over the Indian and Atlantic oceans is still expanding (Fig. 8h). This lag in the response of the Indian and Atlantic sectors was also noted by Navato et al. (1981), Pan and Oort (1983), Wright et al. (1985), and Blazejewski et al. (1986). By JJA +1 the entire global anomaly pattern is significantly weaker (not shown).

4. Discussion and conclusions

Two general conclusions can be drawn from the results of this study. Once concerns the nearly uniform reversal in sign of anomalies during warm events compared to those during cold events. This point is evident in the previous work of van Loon and Shea (1986), Meehl (1987), and Kiladis and van Loon (1988) for

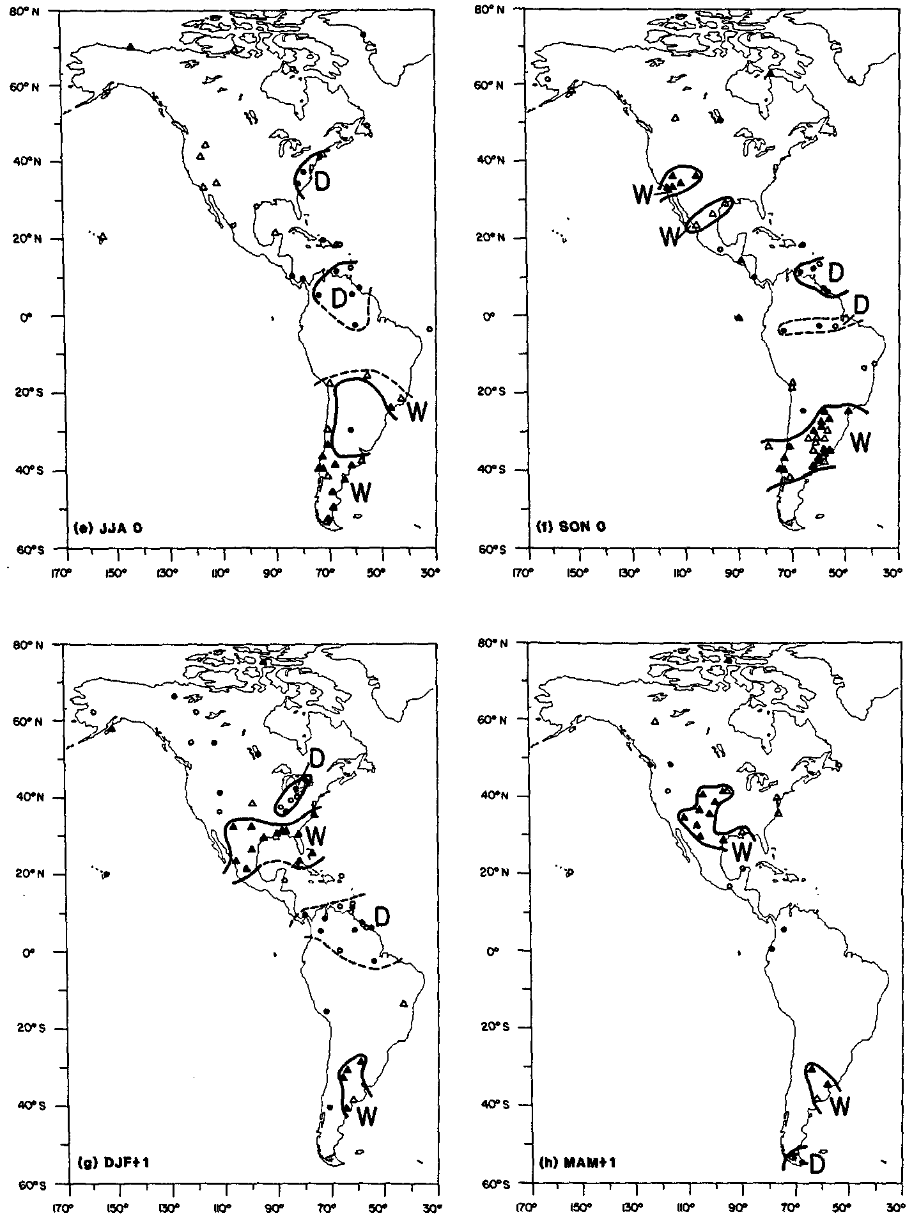

FIG. 5. (*Continued*)

the tropical and subtropical portions of the Pacific and Indian sectors. Other regional studies of the effects of the SO have also reached this conclusion, such as the work of Yarnal and Diaz (1986) on the North American west coast and Rogers (1988) on the Caribbean and tropical South America. Ropelewski and Halpert (1989) obtained similar results for precipitation in 15 of the 19 regions they previously identified with strong warm event signals (Ropelewski and Halpert 1987). The results of the present study confirm that this opposition occurs in most regions of the world with significant climatic signals related to the SO.

The other point relates to the opposition of signals between year −1 and year 0 of both warm and cold events. This should come as no surprise, in view of our above conclusion along with the tendency for warm and cold events to occur in adjacent years in the record. Meehl (1987) pointed out that this phenomenon is a manifestation of the general tendency for cold and warm event conditions over the Pacific sector to alternate in adjacent years (see also van Loon and Shea 1985, 1987; Emery and Hamilton 1985; Gutzler and Harrison 1987; Kiladis and van Loon 1988), and can be interpreted as a modulation of the seasonal cycle.

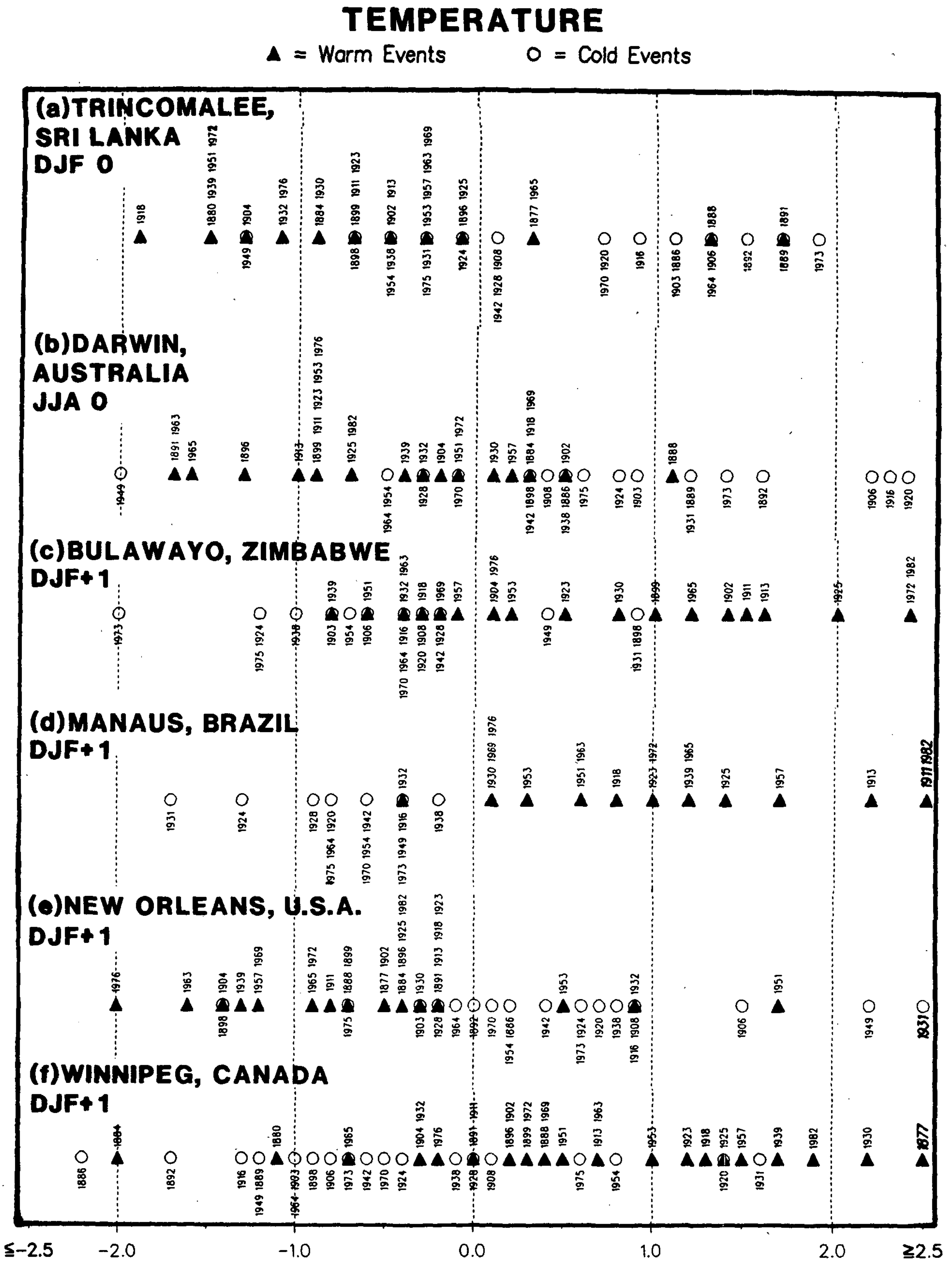

FIG. 6. Plot of normalized departures of individual warm and cold event temperatures at (a) Trincomalee during DJF 0, (b) Darwin during JJA 0, (c) Bulawayo during DJF +1, (d) Manaus during DJF +1, (e) New Orleans during DJF +1, and (f) Winnipeg during DJF +1. Departures are given in standard deviations based on the long-term mean at each station. Warm event departures are represented by triangles with the year 0's of these events listed above them; cold events are represented by circles with their year 0's listed beneath. Years with departures of greater than 2.5 standard deviations are shown in bold type.

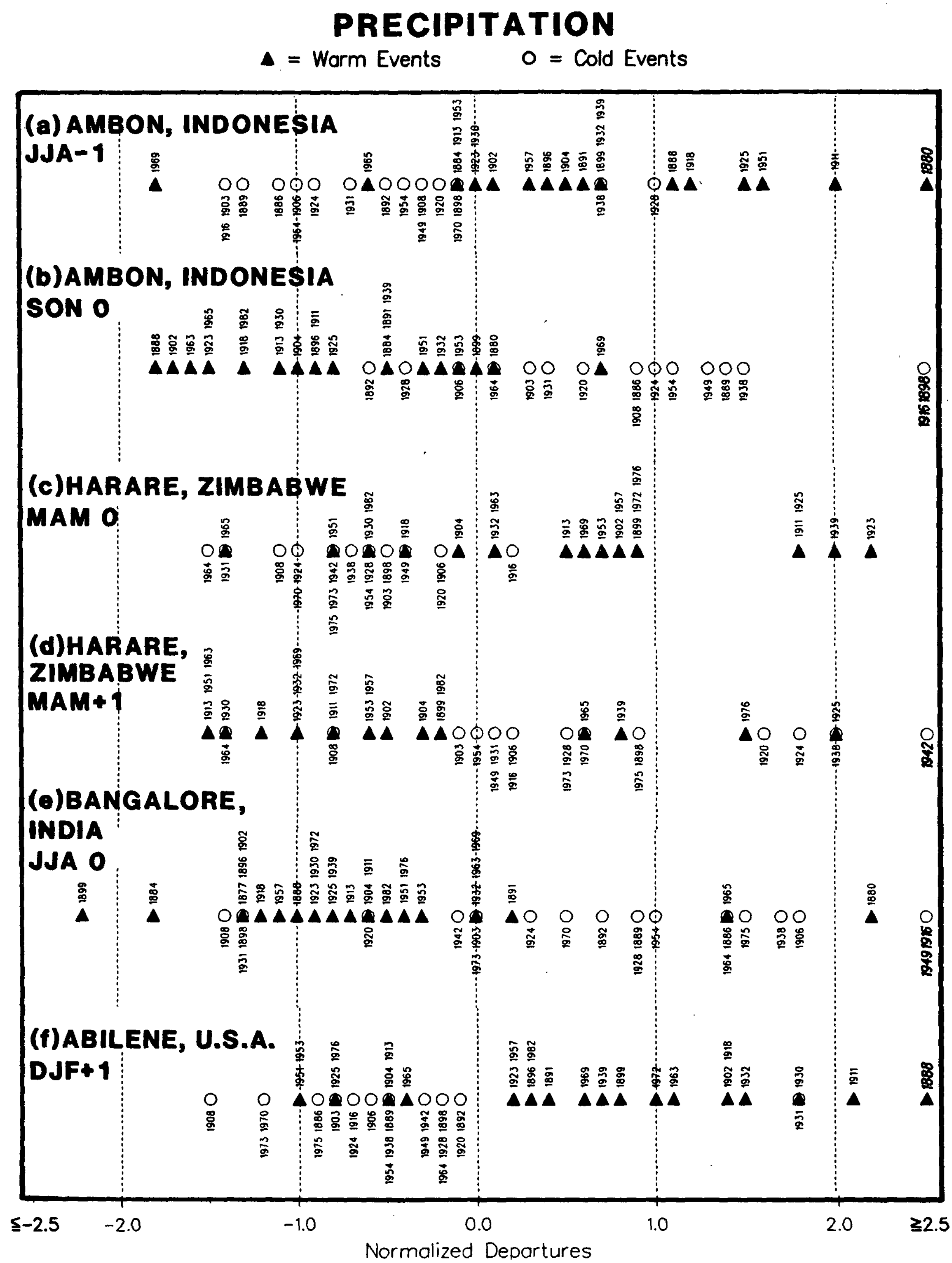

FIG. 7. As in Fig. 6, except for precipitation at Ambon during (a) JJA −1 and (b) SON 0; Harare during (c) MAM 0 and (d) MAM +1; (e) Bangalore during JJA 0; and (f) Abilene during DJF +1.

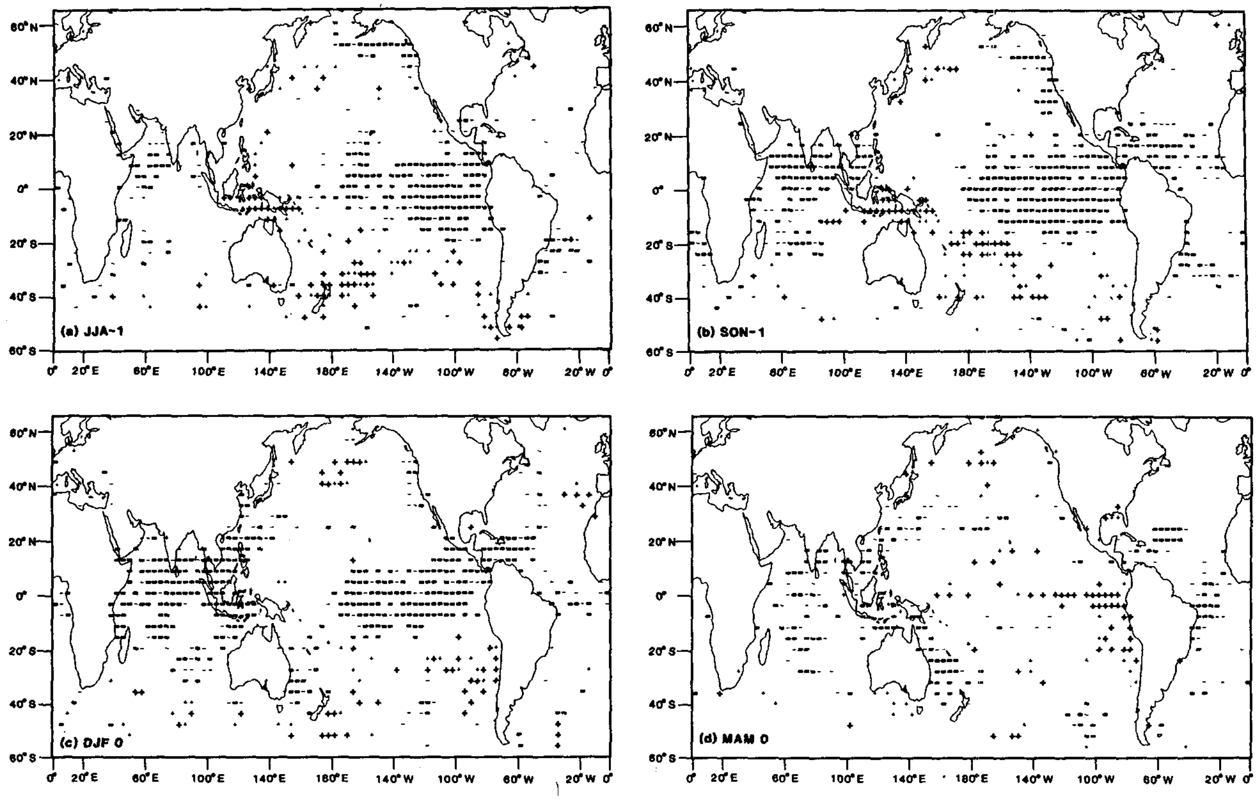

FIG. 8. Maps showing the sign of the composite difference (warm minus cold event) in sea surface temperature for (a) JJA −1, (b) SON −1, (c) DJF 0, (d) MAM 0, (e) JJA 0, (f) SON 0, (g) DJF +1, and (h) MAM +1. Minus (plus) signs denote 4° grid boxes with negative (positive) differences. Unhighlighted symbols are significant at the 90% level, bold symbols are significant at the 95% level.

Thus one extreme of the SO appears to set up the conditions in the ocean/atmosphere system for the transition to the opposite extreme in the following year. While this "biennial tendency" is by no means common to all SO events, it clearly shows up in our composite analysis of global SO signals (see also Lau and Sheu 1988).

A look at Table 1 reveals that, by our classification, in seven cases year −1 of a warm event was the year 0 of a cold event (or, alternatively, seven warm event year 0s were the year +1 of a cold event). Likewise, in nine cases year −1 of a cold event was year 0 of a warm event. Nevertheless, the composite signal is weak by the northern summer of year +1, indicative of the fact that in some instances conditions tend to return more towards "normal" during year +1, while in other cases anomalous conditions persist, resulting in multi-year events. In addition, there are periods in the record in which the SO is relatively quiescent (Trenberth and Shea 1987). Thus the time series of various indices of the SO have a broad spectral peak in the range of 3 to 7 years, with little power at 2 years but a secondary peak at about 28 months (Trenberth 1976; Chen 1982; Rasmusson and Carpenter 1982; Trenberth and Shea 1987). Lau and Sheu (1988) present evidence that fluctuations in the SO may be related to the stratospheric quasi-biennial oscillation (QBO).

We reemphasize that, while many of the signals we have identified are highly statistically significant, there is large inter-event variability associated with many of them, which limits their usefulness for predictive purposes. Nevertheless, it seems worthwhile to advance some hypotheses regarding the origin of some of these anomalies.

By far the strongest and most extensive signal associated with the SO appears as a positive correlation between SST in the eastern equatorial Pacific and surface temperature and SST throughout most of the tropics (see Bradley et al. 1987). The tropical surface temperature and SST signal clearly lags the eastern Pacific signal by about half a year. This relationship has been known for some time (e.g., Newell and Weare 1976; Horel and Wallace 1981; Navato et al. 1981; Hastenrath and Wu 1982; Pan and Oort 1983). It seems probable that tropical air temperature is responding to the change in SST over the tropical oceans, rather than to an advective transport of heat within the atmosphere. If this is true, the problem then becomes one of explaining the observed SST anomaly distribution.

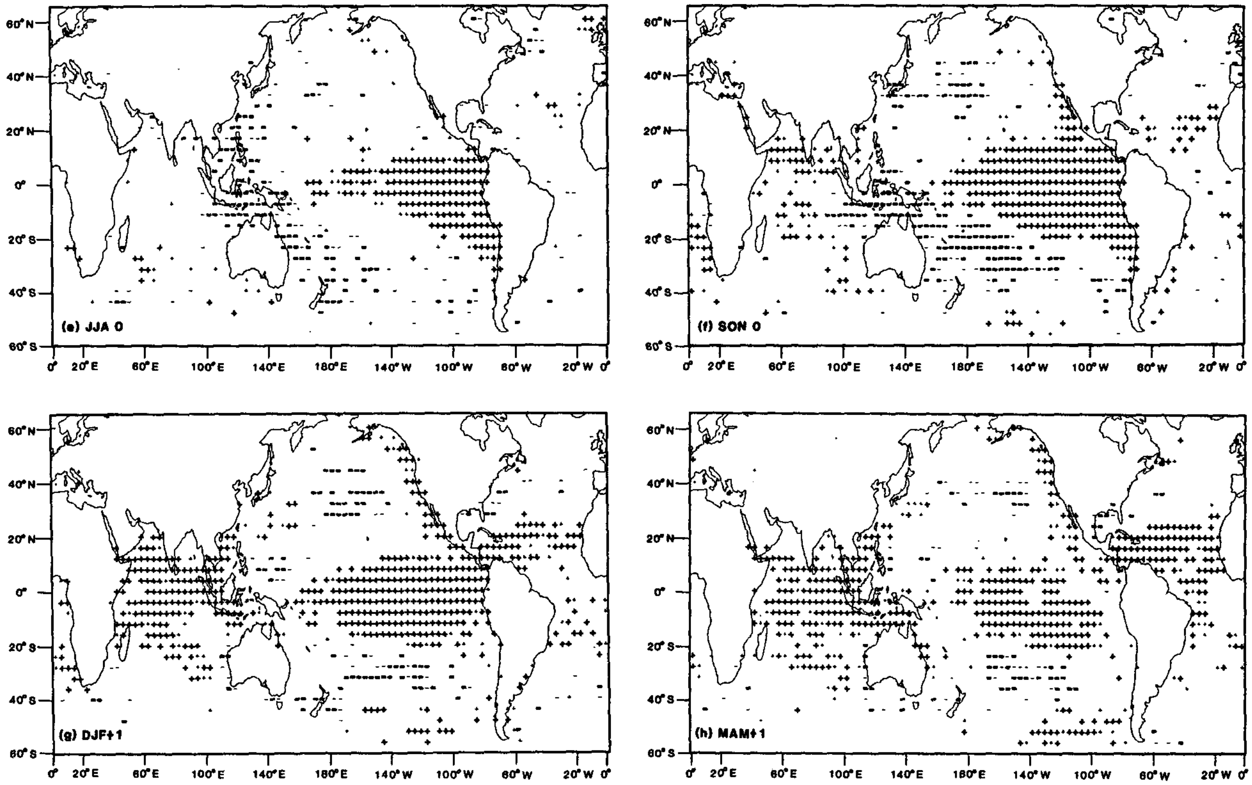

FIG. 8. (*Continued*)

Much recent work has focused on explaining the distribution of SST anomalies in the Pacific sector. SST anomalies in the tropical Pacific respond to anomalous trade winds which modulate the depth of the thermocline and associated upwelling (e.g., Cane 1983; Harrison and Schopf 1984; Philander 1985). Poleward propagation of coastal Kelvin waves and large scale changes in wind stress can account for the observed anomalies along the west coast of the Americas (e.g., Enfield and Allen 1980; Emery and Hamilton 1985), and anomalously strong and cyclonic westerlies appear to be linked to negative SST anomalies in the midlatitude North and South Pacific. In the Indonesian region, enhancement or suppression of the seasonal cycle in wind and cloudiness appear to be instrumental in modifying the SST there (Nicholls 1984a; Hackert and Hastenrath 1986; Meehl 1987).

The far-field in-phase response of the tropical Atlantic and Indian oceans is, however, a more difficult problem (e.g., see Hastenrath and Wu 1982). Advection within the ocean is ruled out here, as well as sensible heating of the ocean by the atmosphere itself. This leaves radiational forcing, due to changes in cloudiness, and local wind stress changes as remaining explanations (Navato et al. 1981). One possibility is that cloudiness over the tropical Atlantic and Indian sectors decreases (increases) during a warm (cold) event in response to changes in heating over the tropical Pacific, through a modulation of both the Walker and Hadley circulations. This is supported by the anomalies in precipitation seen in the tropics during warm and cold events. Overall, large areas of the topics exhibit a decrease in warm event precipitation during DJF +1 and MAM +1 outside the eastern Pacific and coastal South American region affected by anomalously active convection (Figs. 3 and 4). A decrease in wind stress during warm events, again forced remotely by changes in the mass circulation over the Pacific, could also increase the SST locally through a reduction of mixing.

Other strong anomalies associated with the SO affect higher latitudes of the American sector. These signals can be related to circulation changes over the Pacific sector in response to the displacement in convection over the tropical Pacific.

During northern winter North America shows the largest midlatitude response of any land region of the globe. The warm event signal of above normal temperature and below normal precipitation in Alaska and southern Canada and opposite conditions in the southeastern United States fits to some extent the PNA pattern of circulation (Wallace and Gutzler 1981) attributed to forcing by warm event conditions (Horel and Wallace 1981). This would be associated with a strong Aleutian low displaced to the southeast of its

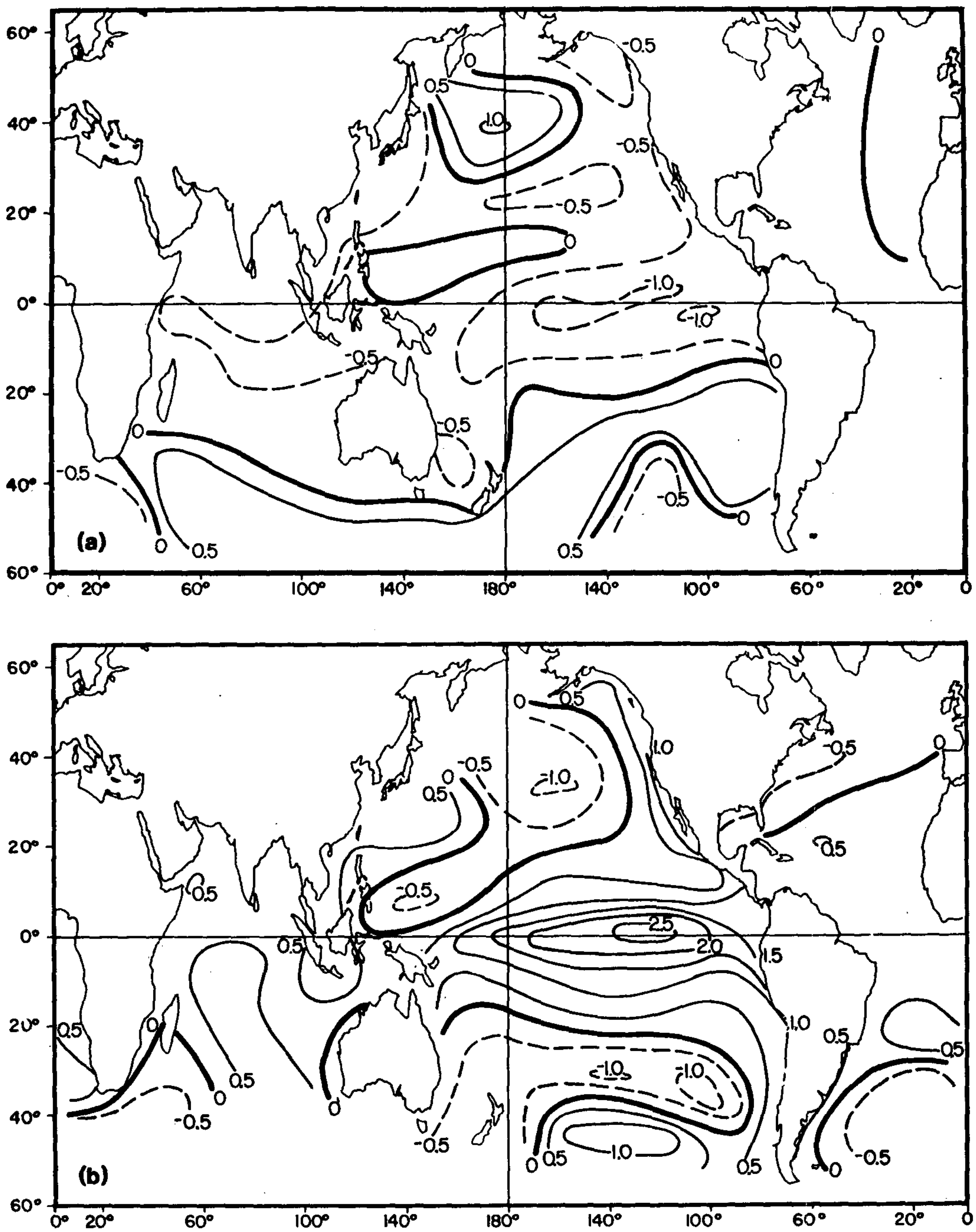

FIG. 9. Subjectively analyzed maps of the composite difference (warm minus cold event) in sea surface temperature for (a) DJF 0, and (b) DJF +1. The contour interval is 0.5°C.

normal position, anomalously strong ridging over northwestern Canada, and a strong subtropical jet and deeper than normal trough extending across Mexico into the Gulf of Mexico. This signal persists into MAM +1, when low-latitude spring storms affect the plains to the east of the Rockies with greater frequency than normal during warm events. A strong subtropical jet is also consistent with above normal temperature and below normal precipitation to its south over Central America during these seasons.

Interpretation of the temperature signal during cold events does appear to support a reverse-PNA pattern during these winters, with strong northerly flow and anomalous cold over western and southern Canada (see Emery and Hamilton 1985; Yarnal and Diaz 1986), and a tendency for ridging with warmth and dryness over the Gulf of Mexico region. The variability in the circulation response over North America to Pacific SST anomalies has been discussed in some detail by Livezy and Mo (1987) and Hamilton (1988).

We have attempted to give a broad view of the temperature and precipitation anomalies over the globe associated with both extremes of the SO. While speculation regarding the forcing of all of these signals is beyond the scope of this paper, we have suggested possibilities for some of them. We hope that this work, together with continued investigation of the local effects of the SO, will stimulate further empirical and theoretical research into the origin and consequences of this global scale ocean–atmosphere phenomenon.

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